

- Deriving ΔS_{Mixing} of ideal gases involves several steps. Write the following steps:
(i) the expression involving S & C_v (ii) the expression involving S & C_p (iii) the expression for S^{Total} (Before Mixing) (iv) the expression for S^{Total} (After Mixing) (v) the expression for ΔS_{Mixing} of ideal gases... [10 marks]
- (i) What are the SI units for $S_{\text{Boltzmann}}$, S_{Standard} , ΔS , ΔS_{Mixing} , R ? [5 marks]
- (i) What is Gibbs-Duhem equation? (ii) What is Gibbs-Chemical Potential expression? (iii) Show Duhem's expressions generated from Gibbs-Chemical Potential expression; (iv) Write a Maxwell relation in partial derivative form [10]
- (i) Write the two expressions {partial & total derivative forms} for the change in free energy for a multi-component closed system; (ii) What is the relation between partial molar entropy of component 'i' and its chemical potential? (iii) Draw qualitatively the chemical potential vs temperature plots for three phases of water (iv) What is the relation between partial molar volume of component 'i' and its chemical potential? (v) Write the expression involving standard chemical potential of component 'i' and the total pressure of the system. [10 marks]
- For a multicomponent system of ideal gases, (i) Write the expression for G^{Total} (Before Mixing) involving chemical potential; (ii) Write the expression for G^{Total} (After Mixing) involving chemical potential; (iii) Write the expression for ΔG_{Mixing} of ideal gases; (iv) How to confirm the expression for ΔG_{Mixing} of ideal gases using a non-chemical potential approach? (v) Why is ΔH_{Mixing} of ideal gases = 0? [10 marks]
- (i) What is Kirchhoff's equation? (ii) What is Hess's Law & how to determine the heat of formation of C_6H_6 from C & H? (iii) What is Nernst Heat Theorem? (iv) Write the Boltzmann entropy equation; (v) What is the relationship between various heats involved in a real gas system? [10 marks]
- (i) Derive $G = H - TS$ & $A = E - TS$; (ii) What is the interpretation of ΔG & ΔA ? (iii) Why the ΔE term is present in the expression for ΔA under constant T? (iv) Why ΔG is negative for spontaneous-irreversible process? (v) Write the Gibbs-Helmholtz relationships; (vi) How Gibbs-Helmholtz relationship connects theory with the practical lab? [15 marks]
- Why $\Delta S_{\text{irr}} > \Delta S_{\text{rev}}$?; (ii) What is the expression for ΔS for a MAIG system involving C_p , T & P; (iii) Which property explains Colligative properties? [10]
- (i) What is the name of the process starting with 'R' which enables skating on the ice? (ii) What is the reason behind ice skating? (iii) Write the mathematical expression [10 marks]
- Write all the Statements of the Second Law of Thermodynamics [10 marks]